Experiment 10.1

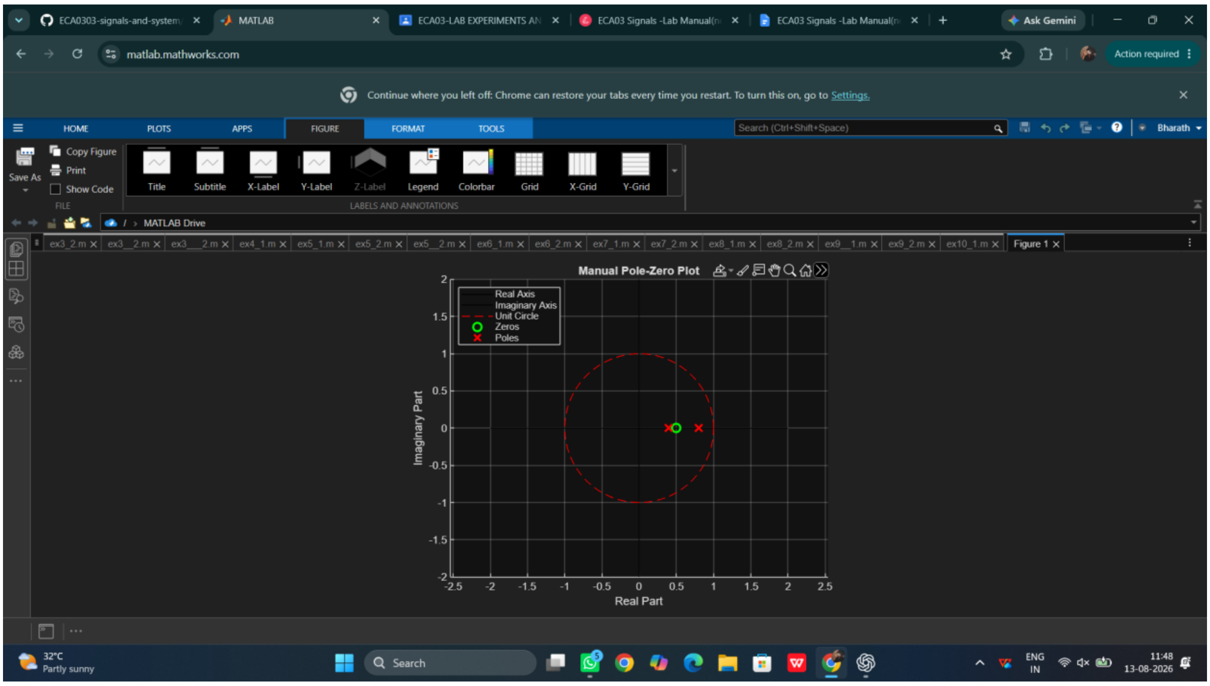
Pole Zero Plot

Program:

```
// Define the coefficients of the transfer function
b = [1, -0.5];
a = [1, -1.2, 0.32];
// Calculate poles and zeros
z = roots(b);
p = roots(a);
// Numerator coefficients (zeros)
// Denominator coefficients (poles)
// Plotting the poles and zeros manually
// Create a new figure
clf;
xgrid();
re = -2:0.01:2;
im = -2:0.01:2;
plot(re, zeros(re), '-k');
plot(zeros(im), im, '-k');
// Plot unit circle
theta = linspace(0, 2*%pi, 100);
unit_circle_x = cos(theta);
unit_circle_y = sin(theta);
plot(unit_circle_x, unit_circle_y, '--r');
// Plot zeros
plot(real(z), imag(z), 'go');
```

```
// Plot poles
plot(real(p), imag(p), 'rx');
// Customize the plot
xlabel('Real Part');
ylabel('Imaginary Part');
title('Manual Pole-Zero Plot');
// Real axis range
// Imaginary axis range
// Real axis
// Imaginary axis
// Calculate zeros
// Calculate poles
// Unit circle
// Zeros as green circles
// Poles as red crosses
legend(["Real Axis", "Imaginary Axis", "Unit Circle", "Zeros", "Poles"],
"location",
"northwest");
xgrid();
```

Output Waveform:



Results:

● Stability: Check the location of the poles relative to the unit circle. If all poles are inside the unit circle, the system is stable; otherwise, it is unstable.

● Frequency Response: The positions of poles and zeros will give insights into the system's behavior in the frequency domain, with poles close to the unit circle potentially leading to high-frequency resonance and zeros affecting the frequency attenuation.

This manual plotting method helps you understand the construction of pole-zero

plots without relying on built-in functions, providing a deeper understanding of the

underlying mathematics in signal processing and control systems.